

HIRADO, Japan

WMO: 478050

Lat: 33.3606N

Lon: 129.5514E

Elev: 192

StdP: 14.59

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	33.8	35.9	17.7	13.5	36.9	19.9	15.1	38.3	22.8	37.5	20.9	38.3	12.5	320	0.404

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.6	87.6	78.8	86.0	78.6	84.3	78.1	80.4	84.6	79.6	83.6	78.9	82.7	8.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.2	152.6	82.6	78.4	148.6	82.0	77.7	145.2	81.5	44.2	85.0	43.4	83.8	42.6	82.8	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
20.0	17.7	15.8	30.6	90.8	2.7	1.8	28.6	92.1	27.0	93.1	25.5	94.1	23.5	95.4	
			DB												
			WB	26.1	81.4	2.1	1.2	24.7	82.3	23.5	83.0	22.3	83.7	20.8	84.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	61.8	44.4	45.7	50.4	57.8	65.0	70.5	77.9	80.5	74.8	66.9	57.6	48.6
	DBStd	12.94	4.90	5.23	4.96	4.36	3.33	3.09	3.62	2.81	3.76	4.09	5.21	5.35
	HDD50	475	184	139	53	3	0	0	0	0	0	0	5	91
	HDD65	2652	639	540	451	218	42	1	0	0	0	27	226	508
	CDD50	4771	10	19	67	237	464	616	866	945	744	523	232	48
	CDD65	1474	0	0	0	2	41	167	401	480	294	85	4	0
	CDH74	9419	0	0	0	1	43	308	2986	4525	1456	100	0	0
	CDH80	1933	0	0	0	0	1	8	571	1162	189	2	0	0
Wind	WSAvg	7.5	8.4	8.2	8.3	7.7	7.1	6.8	7.5	6.8	7.3	7.3	7.1	8.1
Precipitation	PrecAvg	81.4	3.0	3.5	5.3	6.9	7.6	12.6	12.8	9.8	8.1	4.0	4.6	3.2
	PrecMax	117.7	8.5	6.7	9.3	15.3	14.4	21.2	23.9	26.1	20.9	9.6	9.0	8.0
	PrecMin	42.7	0.5	0.9	2.0	2.9	2.3	1.3	2.8	0.9	2.4	0.2	0.4	0.3
	PrecStd	16.9	1.8	1.7	2.0	3.3	3.2	5.4	6.3	6.1	4.6	2.8	2.4	2.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.3	63.2	66.3	71.8	77.4	81.0	88.6	90.6	87.1	79.1	71.9	64.3
		MCWB	57.0	59.1	58.3	60.9	65.2	72.8	78.8	78.8	78.0	73.2	64.9	58.9
	2%	DB	56.4	59.9	63.1	69.2	75.0	78.5	86.5	88.4	84.1	76.5	69.4	60.9
		MCWB	51.7	56.1	57.5	61.2	64.8	72.5	78.7	78.8	76.9	70.2	63.7	55.1
	5%	DB	53.9	56.3	60.8	67.0	73.0	77.0	85.0	87.0	82.1	74.7	67.4	58.6
		MCWB	48.7	51.5	55.9	60.6	64.7	72.6	78.4	78.8	76.0	68.4	61.8	52.9
	10%	DB	51.7	53.6	58.6	65.0	71.3	75.8	83.4	85.6	80.3	73.0	65.6	56.7
		MCWB	46.6	48.3	53.4	59.0	64.2	71.9	78.0	78.5	75.1	66.7	60.0	51.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.1	60.3	62.0	66.8	71.1	76.4	81.4	81.6	79.3	74.9	67.6	61.3
		MCDB	59.9	62.0	63.4	68.9	72.7	78.1	85.1	86.0	83.5	77.8	69.5	63.1
	2%	WB	53.2	56.8	59.0	64.2	68.9	75.3	80.1	80.7	78.3	72.4	65.3	56.8
		MCDB	55.2	59.9	61.9	66.8	71.4	76.8	83.8	85.3	82.3	75.4	68.1	59.3
	5%	WB	49.9	52.8	56.7	62.2	67.4	74.3	79.3	80.0	77.3	70.3	63.2	54.1
		MCDB	53.0	55.2	60.2	65.3	70.5	76.0	82.9	84.3	80.9	73.5	66.5	57.6
	10%	WB	47.1	49.5	54.2	60.2	65.9	73.0	78.5	79.3	76.1	67.9	60.9	51.9
		MCDB	51.0	52.7	57.7	63.4	69.4	74.9	82.0	83.3	79.5	71.6	64.8	56.2
Mean Daily Temperature Range	5% DB	MDBR	8.1	9.2	10.1	10.6	10.7	8.0	7.4	8.6	8.0	8.9	9.4	8.7
		MCDBR	11.2	12.6	13.0	13.8	13.8	10.0	9.9	10.7	10.0	10.3	11.1	11.2
		MCWBR	9.4	10.5	9.9	8.5	7.6	5.5	4.3	4.6	5.1	6.4	8.3	9.2
	5% WB	MCDBR	10.7	11.4	11.9	10.5	10.2	6.6	8.5	9.3	8.6	8.8	9.7	10.5
		MCWBR	10.4	11.6	11.2	8.6	7.7	4.9	4.2	4.5	5.1	6.5	8.7	10.1
Clear-Sky Solar Irradiance	taub	0.442	0.482	0.543	0.562	0.569	0.587	0.568	0.537	0.508	0.465	0.437	0.431	
	taud	2.029	1.959	1.819	1.827	1.869	1.918	2.011	2.108	2.125	2.130	2.129	2.068	
	Ebn at Noon	233	238	234	237	237	232	235	241	242	241	234	228	
	Edh at Noon	44	52	64	67	64	61	56	50	47	43	39	40	
All-Sky Solar Radiation	RadAvg	676	921	1276	1541	1662	1433	1626	1687	1347	1128	808	628	
	RadStd	63	96	89	128	196	179	240	161	133	112	82	44	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (26 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page